

2024 JC2 H2 Physics Preliminary Examination Paper 3 Suggested Solution

1	(a)	Resultant force in any direction must be zero. [B1] Resultant torque about any axis must be zero. [B1]			
	(b)	(i)	Taking moments about the wheel of trailer the sum of clockwise moments = the sum of anti-clockwise moments $F(2.5) = 1900 \times 0.35 \quad [\text{C1}]$ $F = 266 \text{ N} \quad [\text{A1}]$		
		(ii)	Summing all vertical forces $F + F_R = 1900$ $F_R = 1900 - 266 \quad [\text{C1}]$ $F_R = 1630 \text{ N} \quad [\text{A1}]$		
	(c)	Let the horizontal force be F_h . $F_{\text{net}} = ma$ $F_h - 200 = \frac{1900}{9.81}(3.0) \quad [\text{C1}]$ $F_h = 781 \text{ N} \quad [\text{A1}]$			

2	(a)	(lift is perpendicular to the wings)			
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